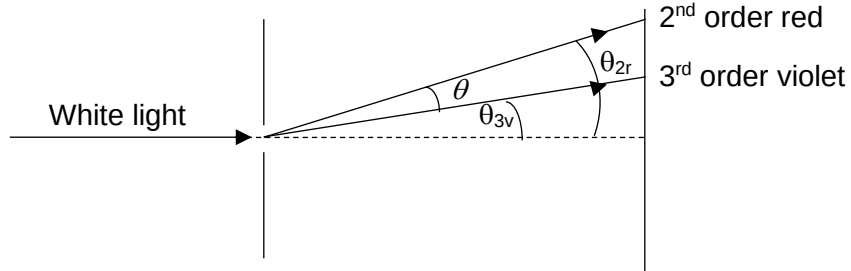


16. **Ans: A**



For each order, red (longest wavelength) will diffract more than violet (shortest wavelength).

2nd and 3rd orders overlap means the 2nd order red diffracts more than 3rd order violet

$$\text{Slit separation } d = \frac{1 \times 10^{-3}}{300} = 3.333 \times 10^{-6} \text{ m}$$

$$\text{Using } \sin \theta = \frac{n\lambda}{d},$$

$$\text{For red: } \sin \theta_{2r} = \frac{2 \times 700 \times 10^{-9}}{3.333 \times 10^{-6}}$$

$$\theta_{2r} = 24.8^\circ$$

$$\text{For violet: } \sin \theta_{3v} = \frac{3 \times 400 \times 10^{-9}}{3.333 \times 10^{-6}}$$

$$\theta_{3v} = 21.1^\circ$$

$$\therefore \theta = \theta_{2r} - \theta_{3v}$$

$$= 24.8 - 21.1$$

$$= 3.7^\circ$$